Investigation of Exfoliated Graphene Flakes and Their Self-Assembly Capabilities to Form a Thin Coating Film

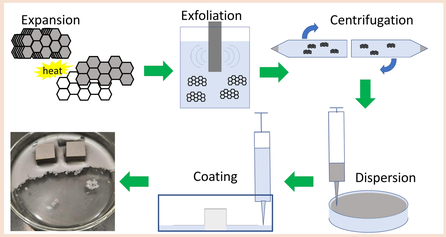
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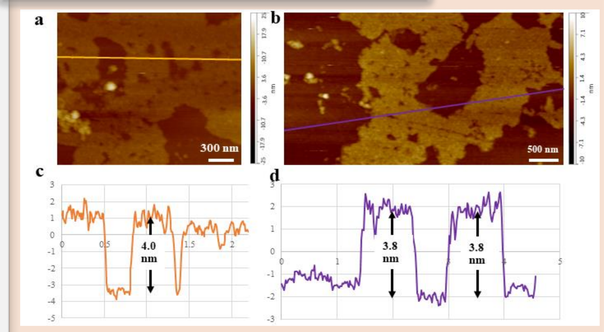
Introduction

Interest in graphene has grown over the last two decades because it is chemically inert, electrically conductive, and has the highest tensile strength of any known material. For these reasons, it is a good candidate for anti-corrosive coatings on metals. In this study, a thermal-expansion liquid-exfoliation method was employed to create multilayer exfoliated graphene flakes (EGFs). A thin film coating was then created through a self-assembly process on an air-water interface. The complementary techniques of SEM, Raman, AFM and laser profile scanning were used to evaluate the qualities of the EGFs for anti-corrosive coatings on additive manufactured steel.



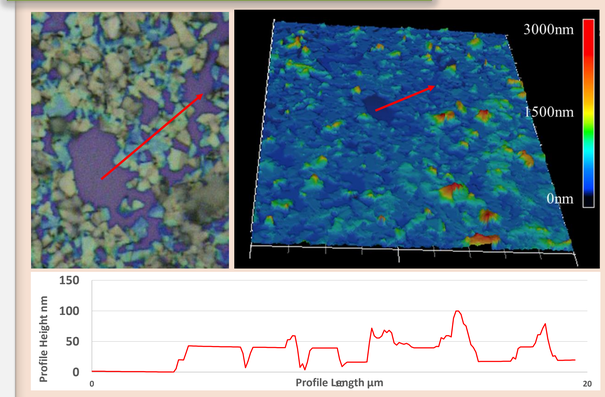
Experimental procedures for EGF coatings following the green arrows: thermal expansion of EG precursor; liquid-phase exfoliation to form EGFs through sonication in NMP; centrifugation and resuspension; dispersion across the water surface to create a film; lowering the water surface to coat an AM cube; image of coating being applied to two AM316 cubes.

Atomic Force Microscopy



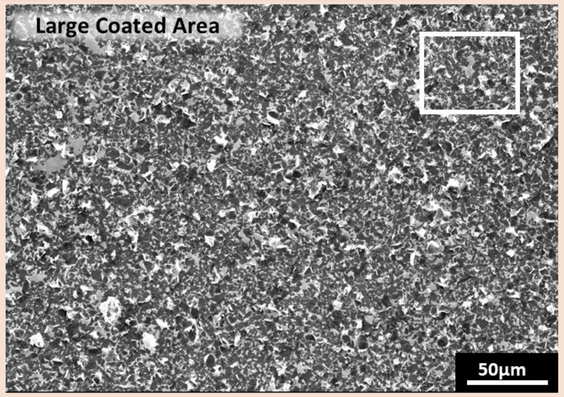
AFM measurements are a reliable method of determining the thickness of graphene flakes and the morphology. Individual thin graphene sheets were identified in the spaces between larger flakes, then analyzed through tapping mode atomic force microscopy. The multilayer flakes in these images are about 4nm in height.

Laser Scanning Microscopy

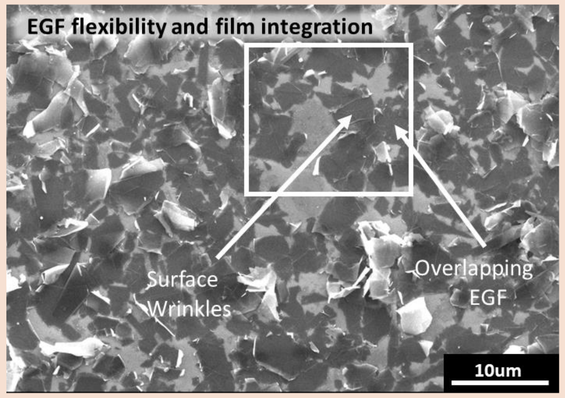


The laser profile was used to determine the overall coating thickness. The lowest points in the scan were 5-20nm, while the average regions from overlapping flakes are typically 60-80nm. The majority of the film is 100nm in height. The tallest peaks are caused by EGFs which are sitting on the film at an angle.

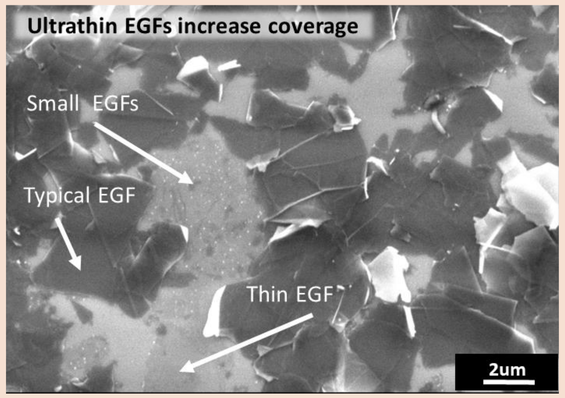
Scanning Electron Microscopy



The EGF coatings were applied to silicon wafers first for characterization. This image shows a single coating over an area of 300um with coverage of ~85%. Large films could coat metal cubes 1cm³ in size. The square in white indicates the area selected for detailed views.



A single exfoliated sheet is highly flexible, shown by the wrinkles on the surface and the flakes overlapping. Van der Waals forces cause the restacking of graphene layers, bonding the flakes together and to the substrate, creating an integrated film.

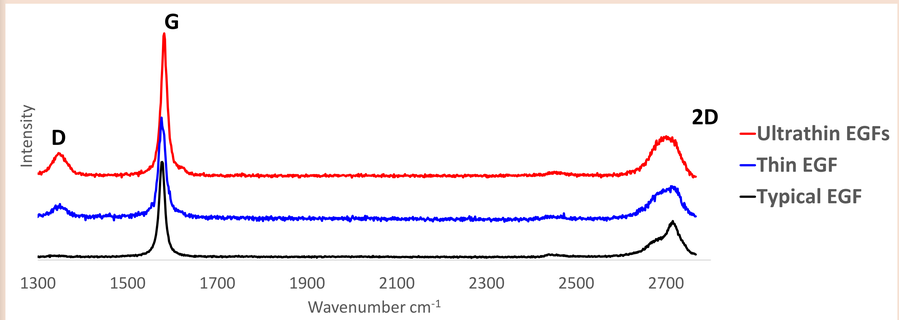


High detail of a gap between larger EGFs. Bare silicon wafer can be seen in the bottom left of the image, next to a thin EGF. Very small and thin EGFs gather in the spaces between larger flakes, increasing coverage.

Raman Spectroscopy & Mapping

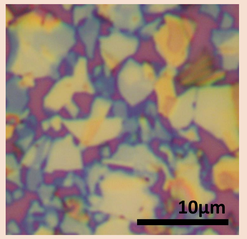
For exfoliated graphene coatings, it is important to know the thickness of the exfoliated graphene flakes and the material defects from the process. Raman spectroscopy can be used to analyze these through characterization of the 3 major graphene peaks; D at 1350, G at 1580 and 2D at 2690.

The D peak is activated by defects in the graphene lattice and from flake edges. The D peak increases for small EGFs because the thinnest pieces have a diameter smaller than the laser spot size (1um). The G and 2D peaks determine the thickness of graphene. As the layer number decreases (thinner graphene) The G peak shifts to higher wavenumbers, the ratio between the 2D and G peak increases, and the shape of the 2D peak changes.



The spectrum in black is representative of the majority of the large EGFs making up the coatings. These flakes are 10-30 layers thick. Any thicker pieces are eliminated by the centrifuging process.

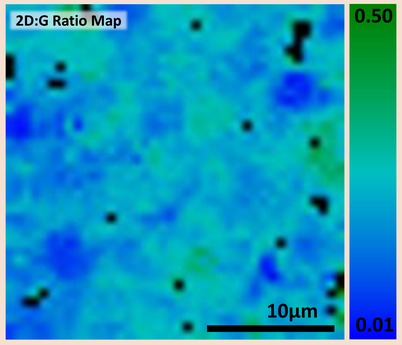
The spectra in red and blue show thin and ultrathin EGFs respectively. The D peak increases from edge effects, the 2D peak becomes stronger and changes shape, and the G peak shifts to a higher wavenumber.

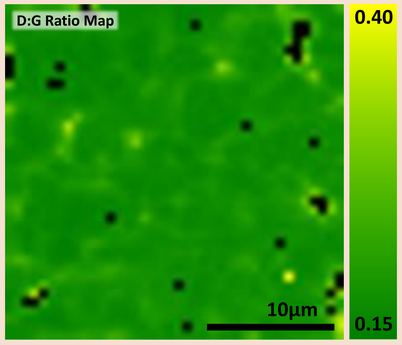


The Raman ratio's for 2D:G or D:G can be mapped to build a picture of the changes in thickness or defects across a film. It can also be used to see the coverage of the film, and to confirm the smallest flakes present in the gaps. Raman mapping was performed on the area shown in the optical reference image to the left.

The map for the 2D:G ratio shows that larger flakes are composed of a range of layers, while thinner flakes collect in the gaps and on edges. The average 2D:G ratio for EGFs is 0.26 and up to 0.5.

The map for the D:G ratio shows that typical EGFs have a minimal D:G ratio of 0.01, with flake edges having a higher ratio. The gaps are filled by the particles with the highest ratio.

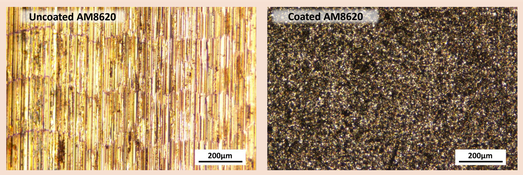




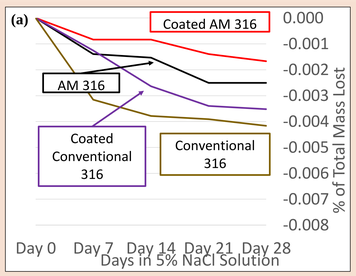
Conclusion & Future Work

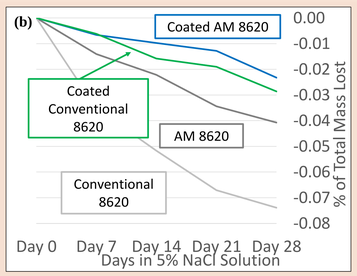
EGF coatings are a promising route to creating a thin film anti-corrosive coating. The complementary techniques of SEM, Raman, AFM and laser scanning microscopy show that the liquid-exfoliation of EGFs result in multilayer graphene with high crystallinity, which can be assembled into coatings of around 100nm thick.

Future work will focus on corrosion testing techniques such as nanoindentation, weight loss and electrochemical testing on conventional and additively manufactured steels to measure the anti-corrosive performance.



Optical images of the as-printed surface of an additive manufactured 8620 steel sample before and after coating. Coatings were applied to conventional, and additive manufactured 316 stainless steel and 8620 mild steel.





Preliminary results from a manuscript in preparation. Weight loss corrosion testing suggests improved anti-corrosive properties of the surface steel samples coated with EGF thin films.

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